

Analysis of optical microwave properties and their use in interferometry

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We investigated the angular distribution of a Gunn-Diode as microwave transmitter and found the full width at half maximum (FWHM) angle to be $\theta_H = (25.7 \pm 1.4)^\circ$. With this diode the focal length of a paraffin wax lens was determined to $f = (39.82 \pm 0.89)$ cm. Furthermore, the wavelength of the used microwaves was measured through standing waves to $\lambda_{SW} = (3.333 \pm 0.015)$ cm and with a Michelson Interferometer to $\lambda_{MI} = (3.280 \pm 0.028)$ cm. This microwave radiation was then used for Bragg diffraction on a cubic crystal model and the lattice constants for the crystal planes (100) and (110) were determined to $a_{100} = (3.942 \pm 0.077)$ cm and $a_{110} = (4.000 \pm 0.051)$ cm. In addition to these quantitative measurements, we also observed frustrated total reflection between two paraffin prisms, investigated the polarization of the emitted microwaves, and confirmed the propagation of microwaves through a waveguide.

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1 Introduction

Electromagnetic waves play a fundamental role in both modern technology and physics. Their interaction with matter gives rise to a wide range of phenomena, including reflection, interference, diffraction, and polarization. These effects can be investigated using microwaves.

Due to their comparatively large wavelengths in the centimeter range, microwaves allow wave phenomena to be observed on macroscopic length scales. This makes them particularly well suited for experimental studies of wave optics and electromagnetic radiation. Moreover, the physical principles underlying these phenomena form the basis of important analytical techniques used throughout modern physics and materials science.

In this work, several fundamental properties of microwave radiation are investigated. The wavelength of a microwave source is determined, and phenomena such as polarization, reflection, and frustrated total reflection are examined. Particular emphasis is placed on the use of interference and diffraction effects to determine physical quantities, culminating in the determination of the lattice constant of a cubic crystal model via Bragg diffraction.

2 Angular distribution of microwave radiation

To study the angular distribution of the emitted microwave radiation a Gunn diode transmitter with an attached horn antenna was used and installed on a goniometer with the effective point of emission being on the rotating axis. The receiver was then rotated while keeping a constant distance to the transmitter and the detected intensity was measured depending on the rotation angle θ . For all following experiments we used a transmitter which, according to the manufacturer's data, emits microwaves with a frequency of 9.35 GHz or wavelength of 3.21 cm. The used voltmeter for all parts was the *Agilent-34405a*.

The measured intensity normalized to 1 for $\theta = 0^\circ$ is plotted in polar coordinates in Figure 1. It shows a strict symmetry around 0° and the intensity decreases as the angle increases. To quantify this angle stretch of the radiation we determined the full width at half maximum (FWHM) angle θ_H , at which only half the intensity at $\theta = 0^\circ$ is measured. With an angular uncertainty of 2° and Gaussian error propagation this results in

$$\theta_H = (25.7 \pm 1.4)^\circ. \quad (1)$$

As expected due to the transmitter being a horn antenna, this confirms that the microwave radiation is

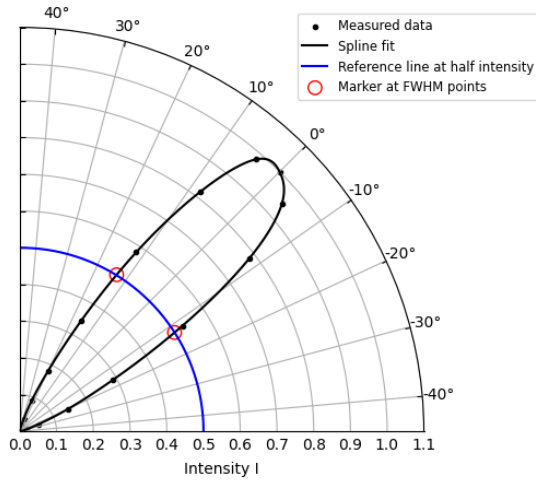


Figure 1: Measured intensity of the microwave radiation (normalized to 1) in dependency of the angle θ . A spline was fitted through the data and the points at the full width at half maximum (FWHM) angle θ_H were marked.

pointed towards the 0° direction.

3 Focal length of a paraffin wax lens

For further investigation of the microwaves, it is necessary to produce parallel rays. To achieve this, wax lenses are used. To determine the focal length of identical wax lenses, a wax lens is placed between the transmitter and the receiver. The distance between the transmitter and the lens is $g = (80.0 \pm 1.4)$ cm. The receiver is placed with maximal possible distance to the lens. Then the radiation is progressively measured for smaller distances. To determine the peak radiation level, a smoothed curve was fitted to the measurement results as shown in Figure 2. The maximum was found at $b = (79.3 \pm 3.5)$ cm, obtained from a third-degree polynomial fit to the measured data. The focal length was then calculated using the thin-lens equation

$$f = \frac{g \cdot b}{g + b} \quad (2)$$

which results in a focal length of $f = (39.82 \pm 0.89)$ cm.

4 Determination of the wavelength via standing waves

We placed the microwave transmitter and a metal plate approximately 1 m apart on the optical bench.

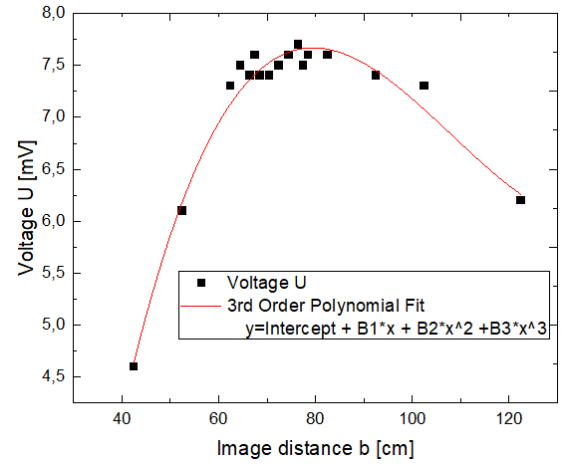


Figure 2: Intensity profile of the microwave radiation on the main beam behind the wax lens for determining the focal length. Plotted is the detector voltage U as a function of the image distance b at a fixed object distance of $g = 80$ cm. The red line shows a fitting curve obtained by a third-degree polynomial fit to determine the intensity maximum. The fit parameters are: Intercept = (-11.2 ± 1.5) , $B_1 = (0.588 \pm 0.060)$, $B_2 = (-0.00586 \pm 0.00077)$ and $B_3 = (1.81 \pm 0.32) \times 10^{-5}$.

The emitted waves are reflected by the metal plate and interfere with the incident waves, thereby forming standing waves. We then used a diode as a receiver to count the number of nodes n in the interval b between transmitter and metal plate. Using the formula

$$\lambda = \frac{2b}{n - 1} \quad (3)$$

we can now determine the wavelength λ of the microwaves emitted by the transmitter. An uncertainty of 0.2 cm was assumed for each interval endpoint resulting in a wavelength of $\lambda = (3.333 \pm 0.015)$ cm. To further verify our calculated wavelength we used an additional experimental setup to determine the wavelength in the next part.

5 Determination of the wavelength via the Michelson interferometer

We use the setup shown in Figure 3. The hardboard plate serves as a semi-transparent mirror used to split the beam. The plate parallel to the transmitter can be moved linearly, while the plate parallel to the detector remains fixed. The two beams are reflected at the metal plates and interfere at the detector. The

relative phase of the two beams depends on the position of the movable metal plate. Constructive interference occurs when the phase difference between beam 1 and beam 2 is an integer multiple of 2π . In contrast, destructive interference occurs when the phase difference is an odd multiple of π , causing the two beams to cancel each other out. By variation of the distance between the transmitter and the opposing metal plate periodic voltage oscillations can be observed at the detector with the distance between two successive minima being equal to $\frac{\lambda}{2}$. Again, a diode was used as a receiver to count the number of nodes n in the interval b . b is the distance between the two positions of the movable metal plate between which we counted the number of observed nodes n . The wavelength can be determined using equation (3) and gives $\lambda = (3.280 \pm 0.028)$ cm. An uncertainty of 0.2 cm was assumed for each interval endpoint. On the label of the microwave-transmitter the frequency f was given as $f = 9.35$ GHz which corresponds to a wavelength of about 3.21 cm using $\lambda = \frac{c}{f}$. For the speed of light $c = 299\,792\,458 \frac{\text{m}}{\text{s}}$ [2] was used. The two experimentally determined values for the microwave wavelength are consistent within their respective uncertainties. Nevertheless, both values exceed the wavelength specified by the label. The deviations can likely be attributed to unwanted reflections within the experimental setup and slight frequency instabilities of the microwave transmitter.

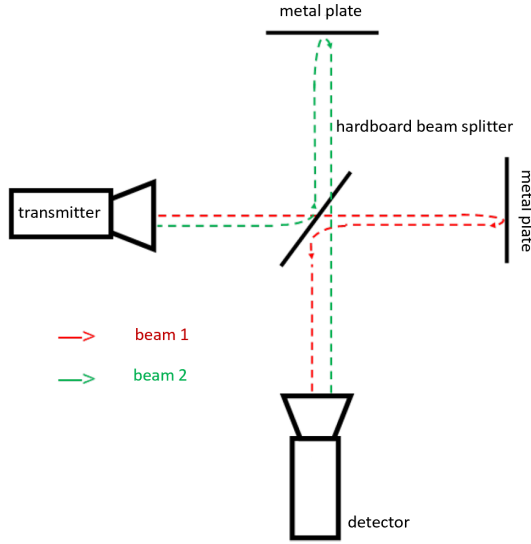


Figure 3: Experimental setup for the determination of the microwave wavelength emitted by the emitter using a Michelson interferometer. The hardboard plate serves as a semi-transparent mirror used to split the beam. The plate parallel to the transmitter can be moved linearly, while the plate parallel to the detector remains fixed.

6 Total reflection and tunnel effect

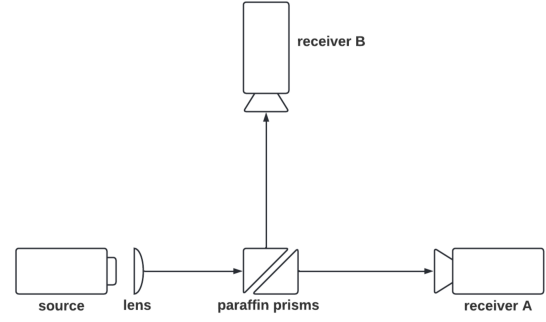


Figure 4: Experimental setup for observing total reflection (adapted from [1]). The source emits microwave radiation at the two paraffin prisms with adjustable gap distance d between them. Receiver A measures the transmitted and receiver B the reflected intensity.

Considering a boundary between two different isotropic media with refractive index n_1 and n_2 , the incident angle α of an incoming electromagnetic wave and the reflected angle β of the outgoing wave are, according to Snell's law [5], related by

$$\frac{\sin \alpha}{\sin \beta} = \frac{n_2}{n_1}. \quad (4)$$

If $n_1 > n_2$ then there is a critical angle α_c for which $\beta = 90^\circ$. For $\alpha \geq \alpha_c$ total reflection occurs. Nevertheless, an evanescent wave still propagates from the surface with an amplitude that decays exponentially with distance from the interface. For the vacuum wavelength λ_0 the intensity follows

$$I_{\text{transmitted}} \propto e^{-\frac{2z}{s}} \quad (5)$$

with the distance z from the surface and

$$s = \frac{\lambda_0 \cos \alpha}{2\pi \sqrt{n_1^2 \sin^2 \alpha - n_2^2}} \quad (6)$$

[1]. This evanescent wave can be captured if a second medium with similar refractive index is brought close to the surface where the total reflection takes place. The totally reflected wave can then be detected in the second medium even if the two media do not touch. This is called frustrated total reflection. This is analogous to the effect of quantum tunneling, where the amplitude of a particle wave doesn't vanish instantly at a potential barrier but decreases exponentially. With the Gunn-Diode as microwave source and two paraffin prisms the effect was observed using the experimental setup shown in Figure 4. The transmitted

and reflected intensities measured by both detectors are plotted logarithmically against the gap distance d between the prisms in Figure 5 together with the total intensity. For $d = 0$ the reflected signal is 0 and not shown in the logarithmic plot, and the transmitted signal is used to normalize all other intensities. As indicated by the linear function fitted to the data points, the transmission intensity follows an approximately exponential decay, which confirms the relation in (5). Simultaneously the measured reflection increases as expected with increasing gap distance. The total intensity remains roughly the same independent of the distance. This corresponds to the theoretical expectation of $R + T = 1$.

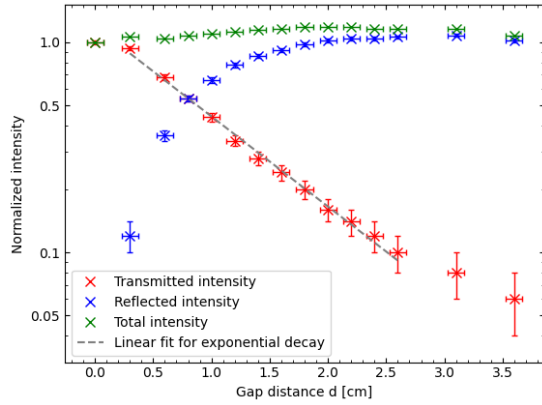


Figure 5: The measured and normalized transmitted, reflected, and total intensities are plotted against the gap distance d between the prisms. The transmitted intensity follows an exponential decay, as indicated by the linear fit in this logarithmic plot.

7 Polarization of the emitted microwaves

To investigate the polarization properties of microwaves, the transmission of a metal grid was measured as a function of the rotation angle over a range from 0° to 180° . In the initial position at 0° (and similarly at 180°), the grid bars are perpendicular to the transmitter and receiver. In this configuration, the detector registers maximum intensity. This can be explained by the fact that the freedom of movement of the electrons across the direction of the rods is severely restricted. Hardly any energy is absorbed or reflected, so that the emitted waves pass through the grid almost unimpeded and the receiver detects a maximum intensity. The pronounced minimum in the measured curve (Figure 6) indicates the successful cancellation of transmission when the rods are oriented parallel to the electric field emitted by the

transmitter. A current flow is induced in the grid rods, which reflects the wave almost completely.

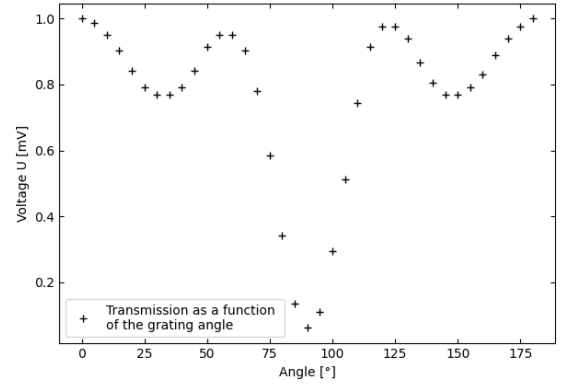


Figure 6: Voltage dependence of the microwave receiver as a function of the polarization grid angle in a 180° measurement series. The graph illustrates the modulation of transmitted intensity due to the orientation of the polarization grid relative to the electric field vector of the electromagnetic wave.

A measurement in which the transmitter and receiver are perpendicular to each other could not be quantitatively evaluated experimentally, since the remaining signal in the detector could no longer be significantly distinguished from the background noise. Theoretically, however, introducing the grating at an angle of 45° in such a crossed configuration leads to a relative intensity maximum. The grating acts as a polarization filter in this case: the vertical field of the transmitter is split at the grating into a parallel and a perpendicular component. The transmitted component oscillates at a 45° angle and thus has a projected field component along the horizontal receiver axis, which can be measured.

8 Determination of lattice constant of a cubic crystal

To analyze a crystal using Bragg reflection, microwaves with a known wavelength, determined in a previous part of the experiment, were used. The condition for Bragg reflection maxima is given by

$$n\lambda = 2d \sin \theta \quad (7)$$

with n being the diffraction order, λ the wavelength of the microwaves, d the spacing between adjacent crystal planes, and θ the angle at which the reflection maximum occurs. The experimental setup is shown in Figure 7. Starting at $\theta = 0^\circ$, the crystal was rotated by an angle θ in steps of 2° , while the detector was rotated by 2θ , ensuring the symmetric condition required for Bragg reflection. The Bragg condition

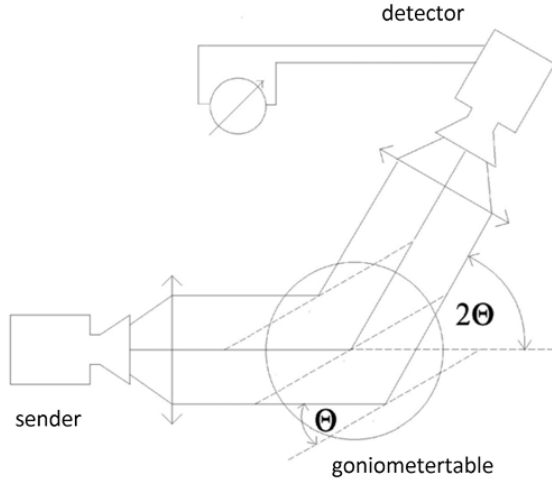


Figure 7: Experimental setup for the determination of the lattice constant of a cubic crystal using Bragg diffraction and microwaves of known wavelength [1].

itself is independent of the orientation of the crystal planes, which are described by the Miller indices h , k and l . For a cubic crystal, the spacing between lattice planes can be expressed as

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \quad (8)$$

where a denotes the lattice constant. Combining both equations enables us to determine the lattice constant a using the equation

$$a = d_{hkl} \sqrt{h^2 + k^2 + l^2} = \frac{n\lambda}{2 \sin \theta} \sqrt{h^2 + k^2 + l^2} \quad (9)$$

As described before, n is the diffraction order. Measurements were performed for the (100)- and (110)-crystal planes with (h,k,l) . These planes are shown in Figure 8. In each case, the corresponding crystal face was positioned parallel to the incident microwave beam on the goniometer table at $\theta = 0^\circ$.

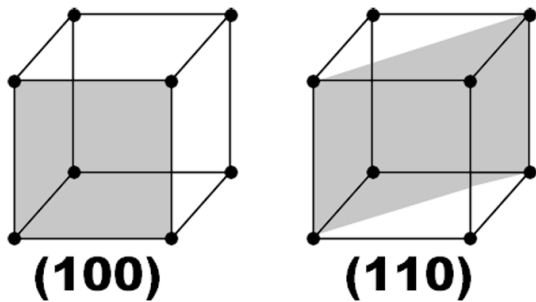


Figure 8: Crystal planes for a cubic crystal with their corresponding Miller indices below [3].

The measured data for the (100)-crystal plane is shown in Figure 9 as a function of the angle θ . The

crystal consists of a Styrofoam-covered cubic lattice. The peak at $\theta = 0^\circ$ is not caused by the crystal itself, but results from the transmitter beam directly reaching the detector. For the (100)-plane we determined the maximum of first order at an angle of $\theta = (24.0 \pm 0.5)^\circ$. The uncertainty was estimated from the graphical reading error.

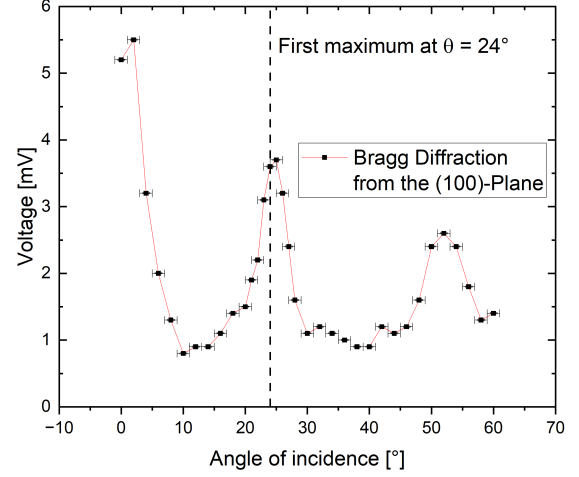


Figure 9: Measured Bragg diffraction from the (100) plane for angles between $\theta = 0^\circ$ and $\theta = 60^\circ$. The first-order maximum was determined graphically and is marked by a dotted line.

Similarly, the measured data for the (110)-crystal plane is shown in Figure 10. Here we determined the maximum of first order at an angle of $\theta = (34.5 \pm 0.5)^\circ$.

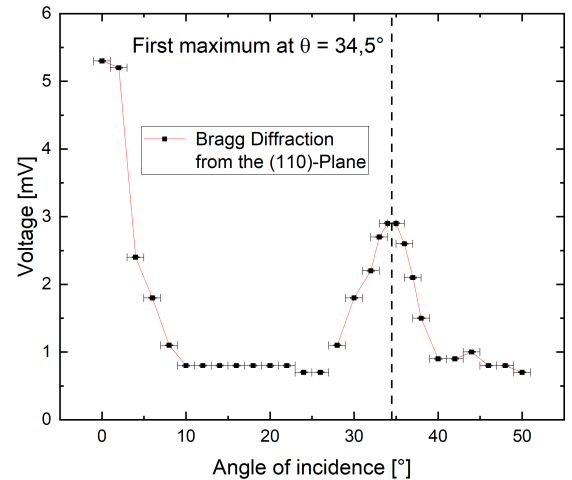


Figure 10: Measured Bragg diffraction from the (110) plane for angles between $\theta = 0^\circ$ and $\theta = 50^\circ$. The first-order maximum was determined graphically and is marked by a dotted line.

Using Equation 9, the lattice constant was deter-

mined from each measured reflection maxima. As wavelength we used $\lambda = 3.21 \text{ cm}$ as derived from the label on the microwave transmitter:

$$(100) : a = (3.942 \pm 0.077) \text{ cm}$$

$$(110) : a = (4.000 \pm 0.051) \text{ cm}$$

Both values are consistent within their uncertainties and are plausible considering the dimensions of the crystal model used in the experiment. The experiment demonstrates that interference effects of electromagnetic waves can be used to investigate structural properties of materials.

9 Waveguide

When an electromagnetic wave is incident on a conducting surface, it is reflected. By introducing additional conducting boundaries, the radiation can be confined to a limited region of space. Such a structure is called a waveguide. The conducting walls impose boundary conditions on the electromagnetic field, allowing only certain field distributions, known as propagation modes, to exist inside the guide. The radiation is therefore guided through the interior of the waveguide rather than propagating freely through space. [4]

To investigate this behavior, the microwave transmitter was placed at one end of a metallic hose and the receiver at the other end. Despite the length and curved shape of the hose, a clear signal was detected at the receiver, demonstrating that the microwaves were successfully guided through the waveguide.

To verify that the radiation propagated through the interior of the hose rather than along its surface, one end of the waveguide was closed with a metal cap. After closing the waveguide, no signal could be detected at the receiver. This confirms that the microwaves were transmitted through the interior of the waveguide.

The polarization of the transmitted radiation was investigated by rotating the receiver. A strong dependence of the detected intensity on the relative orientation of transmitter and receiver was observed, indicating that the microwaves remained linearly polarized after propagation through the waveguide. The observed signal maximum at a receiver orientation rotated by approximately 90° relative to the transmitter suggests that the polarization direction was altered during propagation through the curved waveguide.

10 Summary

The angular distribution of a Gunn-Diode as a microwave transmitter was investigated. The full width at half maximum (FWHM) angle was found to be

$\theta_H = (25.7 \pm 1.4)^\circ$. Then, the focal length of a paraffin wax lens was determined from the measured intensity distribution behind the lens and found to be $f = (39.82 \pm 0.89) \text{ cm}$.

Furthermore, the wavelength of the used microwaves was measured through standing waves to be $\lambda_{SW} = (3.333 \pm 0.015) \text{ cm}$ and via a Michelson Interferometer to be $\lambda_{MI} = (3.280 \pm 0.028) \text{ cm}$. This microwave radiation was used for Bragg diffraction on a cubic crystal model, and the lattice constants for the crystal planes were determined to be $a_{100} = (3.942 \pm 0.077) \text{ cm}$ and $a_{110} = (4.000 \pm 0.051) \text{ cm}$.

In addition to these quantitative measurements, frustrated total reflection between two paraffin prisms was observed. The transmitted intensity was found to decrease approximately exponentially with increasing prism separation, while the reflected intensity increased correspondingly. Furthermore, the polarization properties of the emitted microwaves were investigated using a metal grid polarizer, and the propagation of microwaves through a waveguide was demonstrated.

References

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